

Thermonuclear Deflagration Flames in Type Ia Supernovae

A 1D Multiparametric Study on the Phlegethon Program

Soutenance de stage, Projet de Recherche

Alexis Spaeth-Lemarchand

HITS, Physics of Stellar Objects group

Tuteurs : Prof. Dr. Friedrich K. Röpke, Dr. Giovanni Leidi, Dr. Alexander Holas

Context: Why Type Ia Supernovae Matter

- Standardizable cosmological candles: measured the accelerated expansion of the Universe
- Thermonuclear explosion of a carbon-oxygen white dwarf near the Chandrasekhar mass
- Precision cosmology now limited by **systematic** uncertainty in the explosion mechanism itself, not by statistics

Improving the physics of the explosion improves our measurement of the Universe

Where the Laminar Flame Speed Fits In

- The explosion is ultimately driven by **turbulent** flame acceleration in 3D: not something this 1D study models directly
- But turbulence itself is shaped by the **local** laminar flame speed: Holas & Röpke (2026) show turbulence suppression is highly localized through the Y_e -dependence of v_l
- v_l is a necessary microphysical input to the larger-scale, turbulent models: not the whole story, but the piece this project addresses

$$v_l = f(\rho, X_i, Y_e)$$

The Gap in the Literature

Timmes & Woosley (1992)	Schwab et al. (2020)
$v_l(\rho, X_i)$	$v_l(\rho, Y_e)$
No Y_e dependence	No composition dependence

Nobody has combined all three parameters into one formula

Goal of this internship: explore the full (ρ, X_i, Y_e) space with Phlegethon, and work towards that combined formula.

The Tool: Phlegethon

A compressible, multiphysics stellar hydrodynamics code Leidi et al. (2026), finite-volume, solving:

$$\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F}(U) = S(U)$$

$$U = \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \\ \rho X_k \end{pmatrix}$$

One equation, one code: compressible fluid dynamics, degenerate-matter thermodynamics, and thermonuclear reaction networks, all coupled together.

Nuclear Networks: More Physics, More Cost

- Progression over the internship: $7 \rightarrow 15 \rightarrow 35$ species
- 56-species network still fails: dense-matrix linear algebra scales as $O(n^3)$ in network size
- Larger networks $\rightarrow$ longer simulations $\rightarrow$ needed to understand and improve parallelization

MPI ranks	OMP threads	wct ($\times 10^2$ s)	Duration
64	1	0.575	1min 22s
64	2	0.323-0.564 (var.)	N/A
64	4	0.554	1min 20s

64	8	0.977	2min 18s
64	16	1.320	3min 06s

- Tested OpenMP scalability, contributed to a dedicated 1D geometry branch of Phlegethon's own source code

A Physical Attractor

$$\mathbf{F}(U) = \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I} \\ (E + P) \mathbf{u} \\ \rho X_k \mathbf{u} \end{pmatrix}, \quad S(U) = \underbrace{S_{\text{grav}}}_{\text{gravity}} + \underbrace{S_{\text{diff}}}_{\text{conduction}} + \underbrace{S_{\text{nuc}}}_{\text{nuclear burning}}$$

Convergence tests on grid resolution, box size, ignition temperature, ignition spot size:

The transient changes. The asymptotic flame speed never does.

→ this motivated the whole simulation design that follows: scale every simulation to its own expected physics rather than using one fixed setup for the entire parameter space.

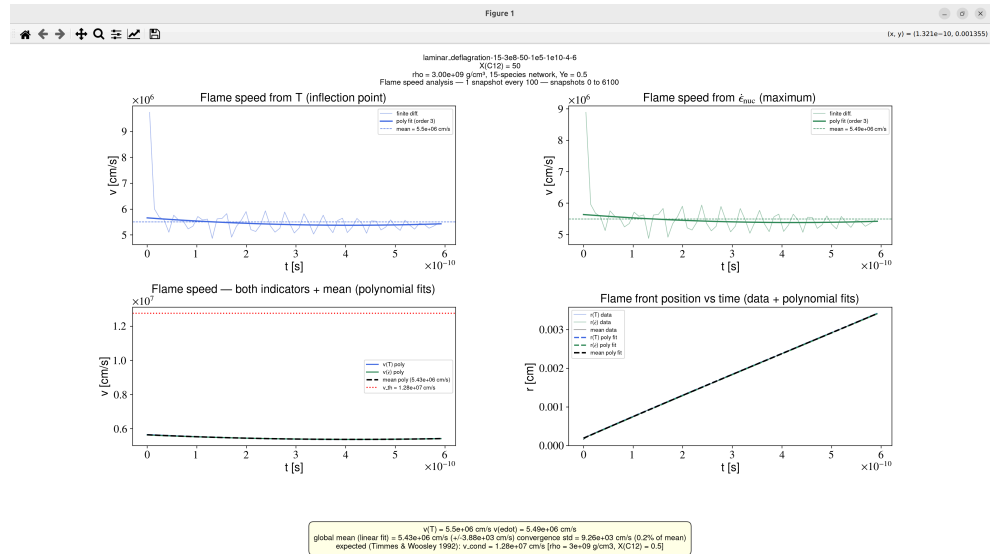
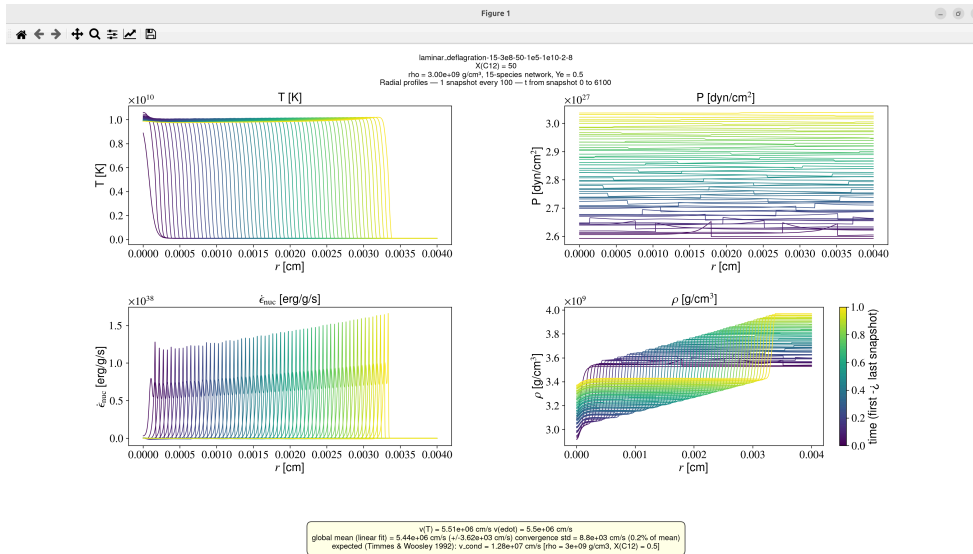
Setup: Self-Similar Simulations, Variable Y_e

- Box length, resolution, and duration all scaled from the flame width and speed predicted by Timmes & Woosley, keeping computational cost roughly constant across the whole phase space
- Varying Y_e without a real neutron-rich isotope: Ne40 spectator species

$$X_s(\text{Ne40}) = \frac{0.5 - Y_e}{0.25}$$

- Ne40 chosen over Ne22: $\sim 6 \times$ less mass fraction needed for the same Y_e shift, perturbing the fuel composition far less

Measuring a Flame Speed



- Front position from two indicators: temperature midpoint crossing + peak nuclear energy release

- Sub-cell resolution via parabolic interpolation

Front Detection Had to Evolve

Thermonuclear Deflagration Flames in Type Ia Supernovae

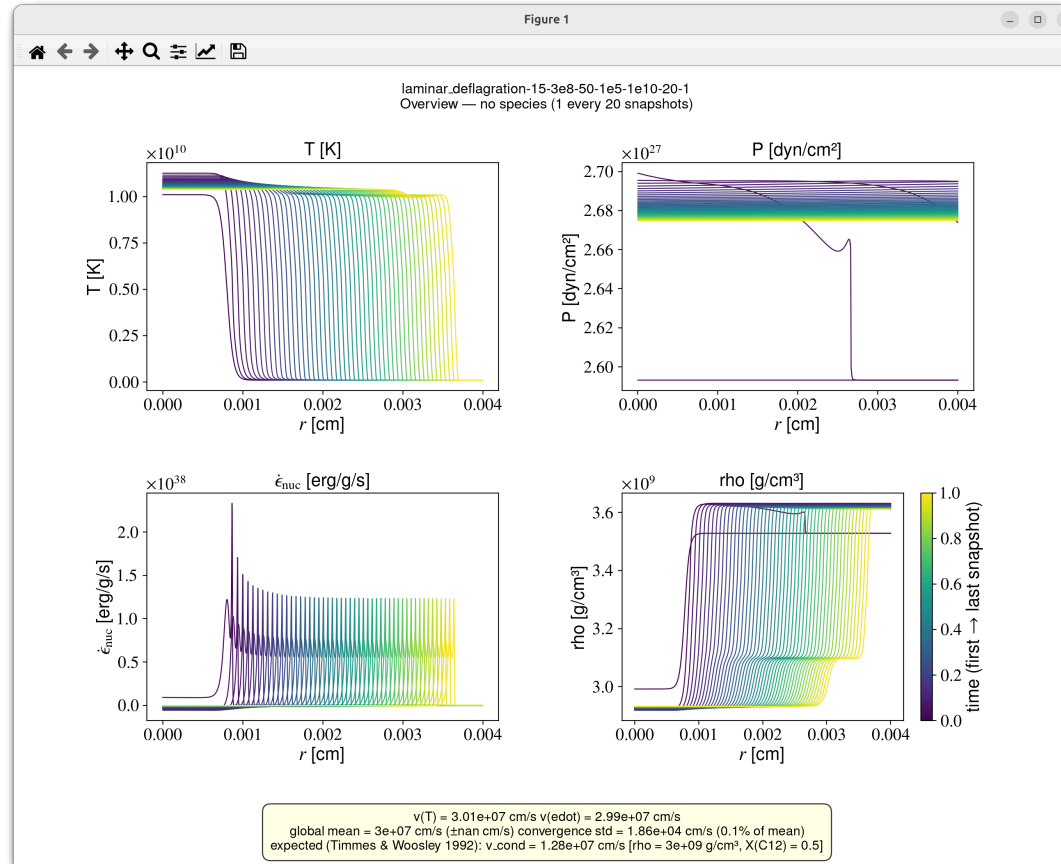


Figure 1: An early failure mode: the inflection-point criterion misidentifies the front.

Replaced by the midpoint-crossing criterion: more robust, requires only that fuel and ash plateaus be identifiable.

Extracting a Speed (1/4): the Naive Average

Thermonuclear Deflagration Flames in Type Ia Supernovae

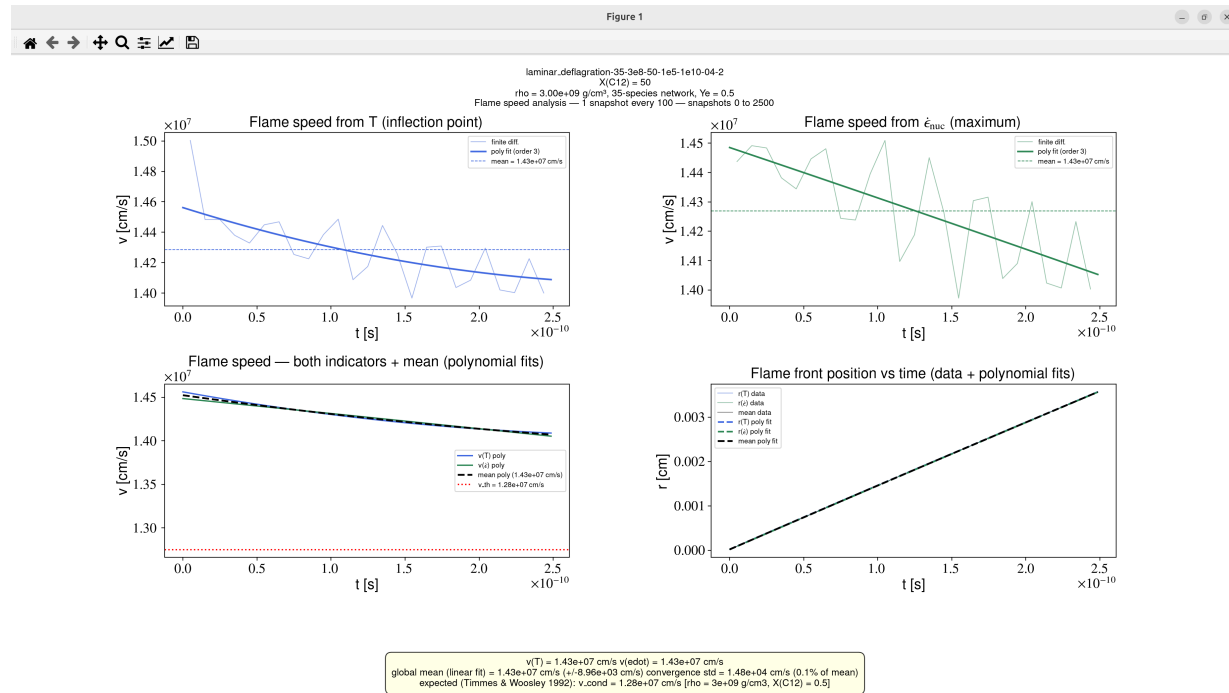


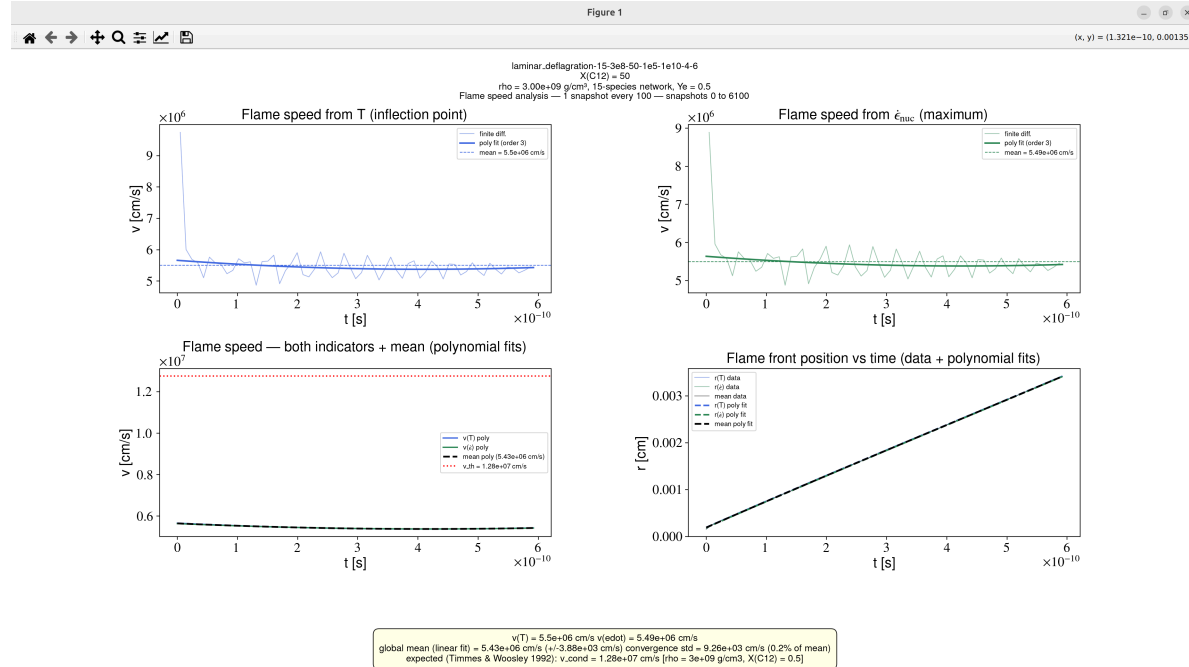
Figure 2: The dashed “mean” line in each panel is the naive time-averaged velocity.

- Average the instantaneous velocity over the whole run

- Biased whenever the transient has not fully decayed by t_{max} , which is the common case

→ needed something smoother, and something that does not silently average over a transient that has not converged.

Extracting a Speed (2/4): the Polynomial Fit



- Low-order polynomial fit to $r(t)$, differentiated analytically

- Numerically robust, but no physical basis for extrapolating to the true asymptotic speed

Extracting a Speed (3/4): the Exponential Fit Can Fail

Thermonuclear Deflagration Flames in Type Ia Supernovae

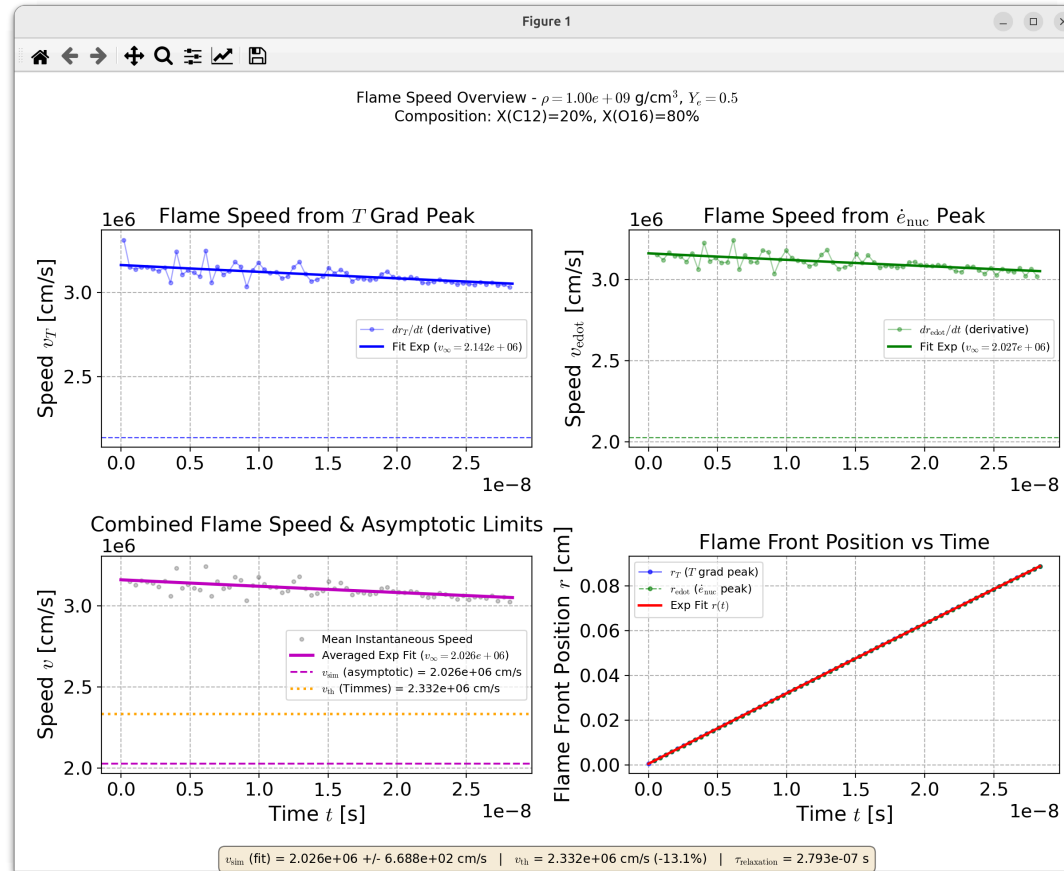


Figure 4: A degenerate fit: the extrapolated asymptote lies far outside the range the data actually cover.

Physically motivated by the classical theory of relaxation toward a stable traveling wave Fife & McLeod (1977); Zel'dovich & Frank-Kamenetskii (1938), but the 3-parameter nonlinear fit is unstable when the run does not cover several relaxation times τ .

Extracting a Speed (4/4): When It Works

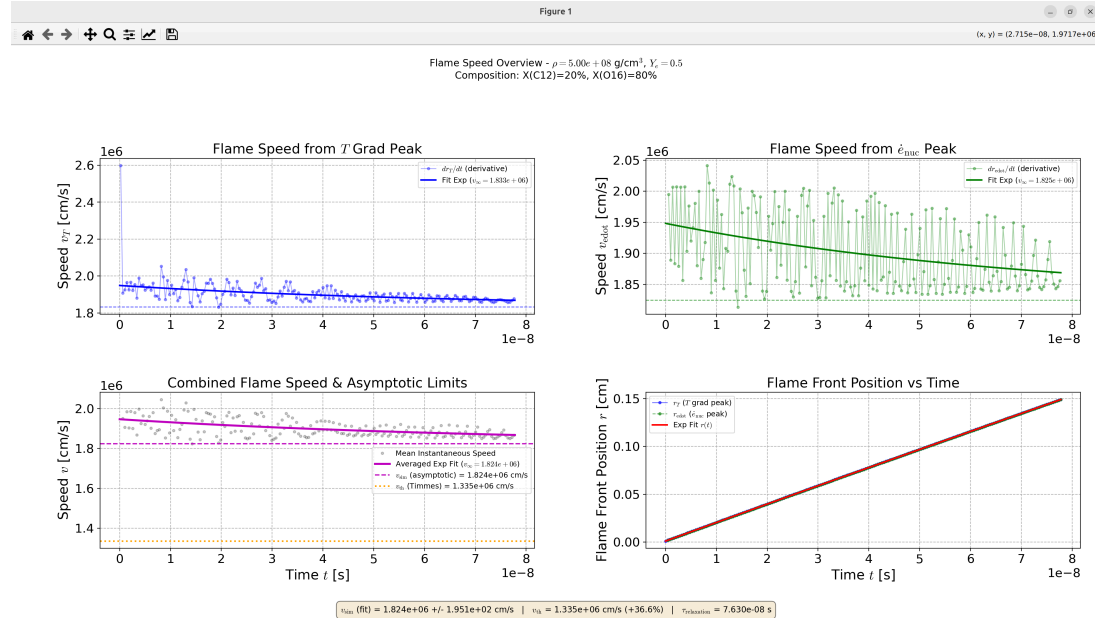


Figure 5: A well-conditioned fit: τ is well resolved within $t_{\max}$.

Fix: a robust, non-parametric cross-check (sliding-window regression + Aitken Δ^2 extrapolation) that flags disagreement instead of trusting a single fit blindly.

Doing Science on Noisy Data

“Change a number and rerun” was never the method: form a hypothesis, design a test, fix the cause

- **Hypothesis:** the extraction noise is partly acoustic waves bouncing across the domain
- **Test:** does the oscillation period scale with box length as expected?
Confirmed, motivating the self-similar box design itself

When the Bug Isn't in the Physics

- Comparison plots showed jumps between simulations that made no physical sense
- Root cause: Y_e was silently defaulting to 0.5 for **every** simulation in the post-processing pipeline
- Fixed by recomputing Y_e directly from the Ne40 mass fraction on disk

One example among several of the same discipline: an anomaly is a question to be answered, not a value to patch around.

Current Results

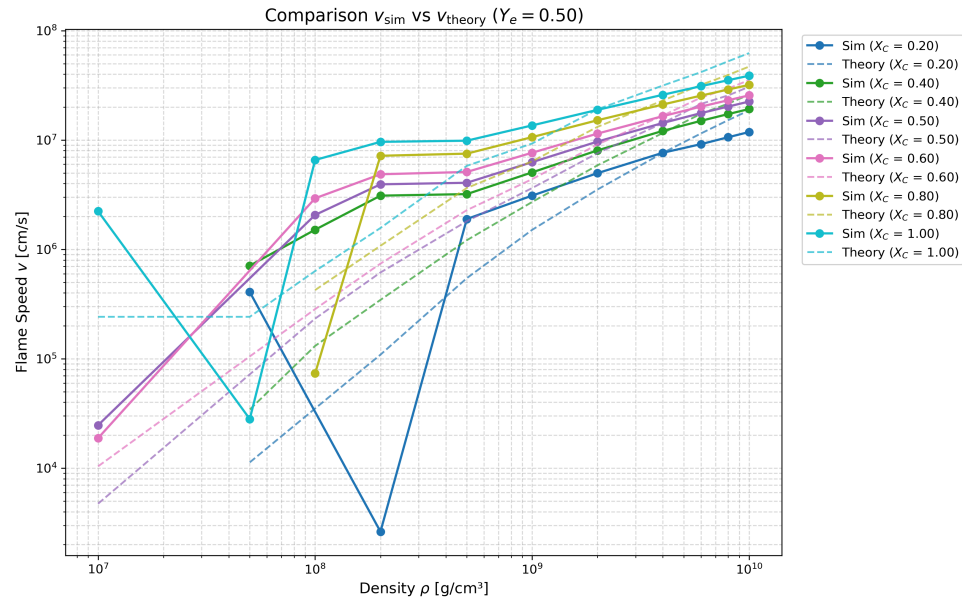


Figure 6: Polynomial fit

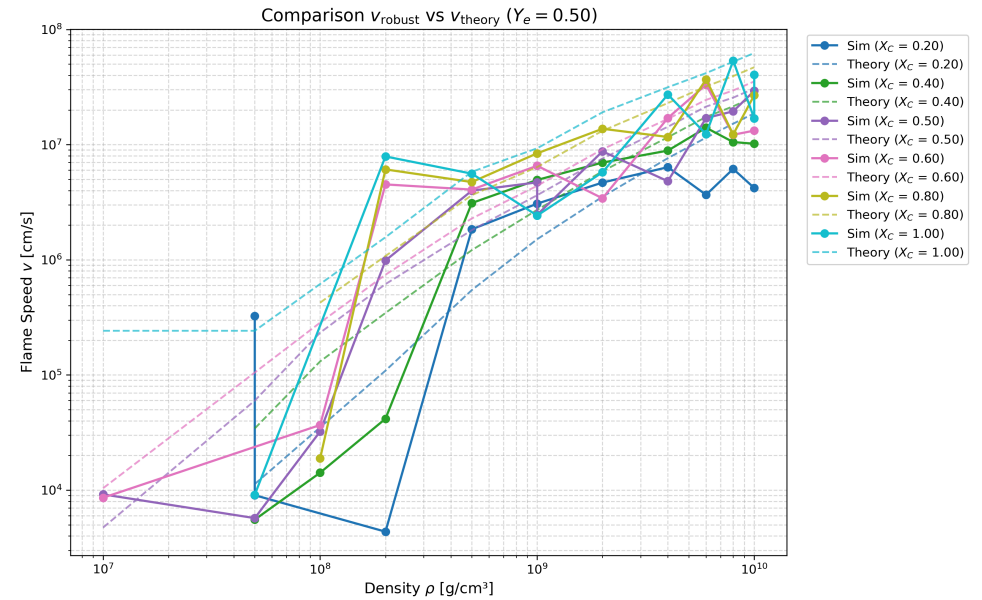
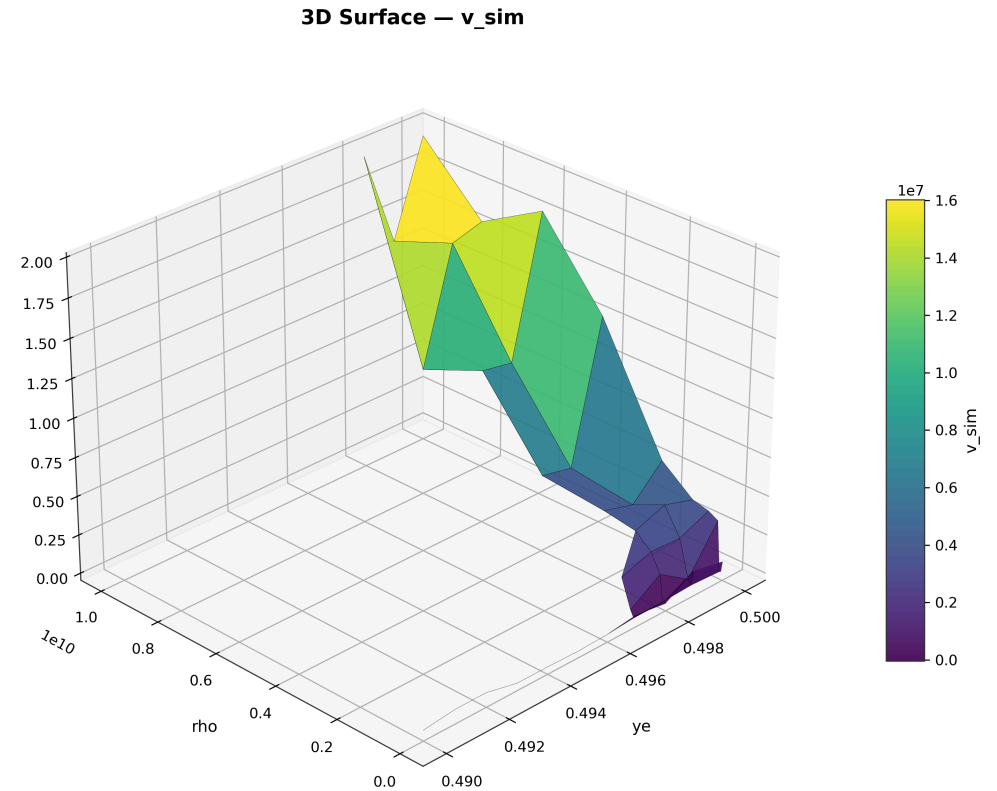
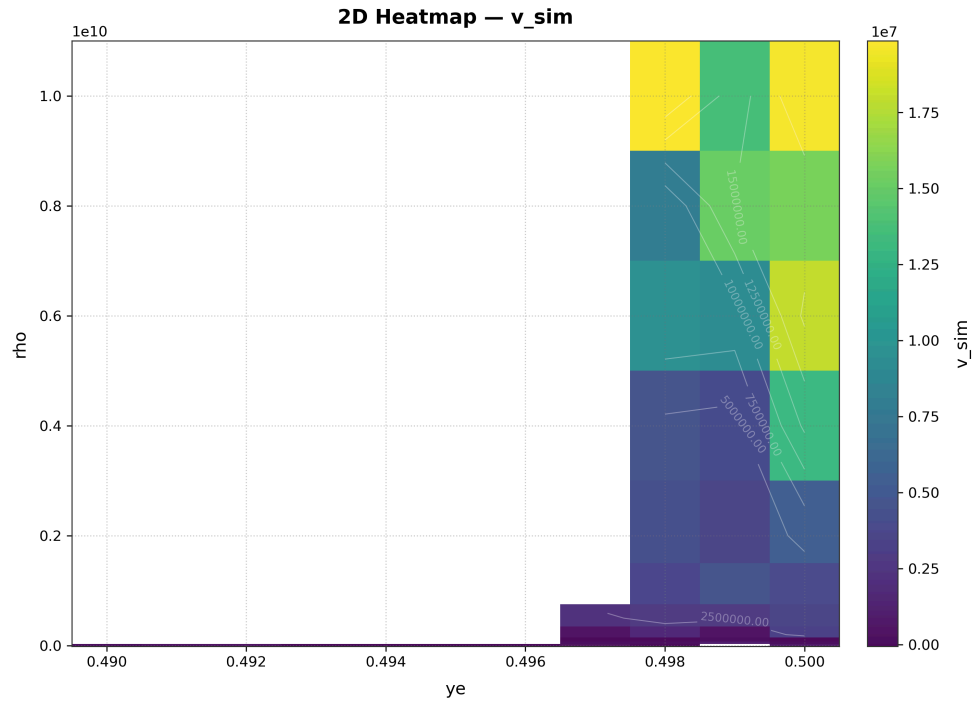


Figure 7: Exponential fit + robust cross-check

- Good agreement with Timmes & Woosley (1992) for $\rho \gtrsim 5 \times 10^8 \text{ g/cm}^3$
- Residual low-density discrepancy, diagnosed (not mysterious): those runs are the ones most starved of resolved relaxation time relative to τ

Phase-Space Maps



$v(\rho, X_C)$ heatmap and 3D surface at $Y_e = 0.5$: produced, but not yet conclusive, since the low-density artifacts above dominate their appearance at low ρ .

Conclusion

Delivered: a validated, self-similar simulation pipeline; a robust flame-speed extraction method; a contribution to Phlegethon's own codebase; a working Y_e -variation methodology.

Not yet delivered: the combined formula $v_l = f(\rho, X_i, Y_e)$, still limited by the low-density transient and an unswept Y_e dimension.

The gap between a simulation producing a number and that number being trustworthy was the real subject of this internship.

Perspectives

- Characterize the τ -density scaling to fix the low-density regime
- Complete the Y_e sweep
- Derive the combined fitting formula
- Sparse-matrix reformulation to unlock the 56-species network

Merci.

References

- Fife, P. C., & McLeod, J. B. (1977). The approach of solutions of nonlinear diffusion equations to travelling front solutions. *Archive for Rational Mechanics and Analysis*, 65, 335–361. <https://doi.org/10.1007/BF00250432>
- Holas, A., & Röpke, F. K. (2026). Drawing the line between explosion and collapse in electron-capture supernovae: I. Impact of conductive flame speed and ignition conditions on the explosion mechanism. *Astronomy & Astrophysics*.
- Leidi, G., Holas, A., Vitovsky, K., Rizzuti, F., Roy, A., Reichert, J., Bayer, K., Gagnier, D., Andrassy, R., Christians, P., Edelmann, P. V. F., Varma, V., Hirschi, R., & Röpke, F. K. (2026). *Phlegethon: a fully compressible magnetohydrodynamic code for simulations in stellar astrophysics*. <https://arxiv.org/abs/2604.12672>
- Zel'dovich, Y. B., & Frank-Kamenetskii, D. A. (1938). The theory of thermal propagation of flames. *Zhurnal Fizicheskoi Khimii*, 12, 100–105.